

Class 33: A Few More Things

Name
Date
Arithmetic

Perform the following operations.

- | | |
|---------------------------|----------------------------|
| ① 6×10 | ② 1000×7 |
| ③ 2×100 | ④ $1,000,000 \times 3$ |
| ⑤ 100×800 | ⑥ $40 \times 10,000$ |
| ⑦ 70×100 | ⑧ 1000×600 |
| ⑨ $300 \times 4,000$ | ⑩ 50×700 |
| ⑪ $800 \times 50,000$ | ⑫ 900×200 |
| ⑬ $400 \times 6,000$ | ⑭ $700 \times 90,000$ |
| ⑮ 5.68×100 | ⑯ $75.4 \times 100,000$ |
| ⑰ 0.3215×10 | ⑱ $873.40011 \times 1,000$ |
| ⑲ 0.001×100 | ⑳ $573.1981 \div 100$ |
| ㉑ $3,678.4 \div 1,000$ | ㉒ $877.053 \div 10$ |
| ㉓ $36.4 \div 10,000$ | ㉔ $1.04 \div 100,000$ |
| ㉕ $5,000 \div 100$ | ㉖ $7,000 \div 1000$ |
| ㉗ $8,000,000 \div 10,000$ | ㉘ $6,000 \div 100,000,000$ |

More Rules/Properties

- I) The Commutative Rule of Addition: $A+B=B+A$
- II) The Commutative Rule of Multiplication: $A \times B = B \times A$
- III) The Associative Rule of Addition: $(A+B)+C = A+(B+C)$
- IV) The Associative Rule of Multiplication: $(A \times B) \times C = A \times (B \times C)$
- V) The Identity Rule of Addition: $A+0=A$
- VI) The Identity Rule of Multiplication: $A \times 1=A$
- VII) The Zero Rule of Multiplication: $A \times 0=0$
- VIII) The Inverse Rule of Multiplication: $A \times \frac{1}{A} = 1$

Name the rule demonstrated below.

- ⑲ $3+0=3$ ⑳ $7+1=1+7$ ㉑ $4 \times 1=4$
- ㉒ $(5 \times 4) \times 2 = 5 \times (4 \times 2)$ ㉓ $6+2=2+6$ ㉔ $0=100 \times 0$
- ㉕ $4 \times \frac{1}{4} = 1$ ㉖ $7+(8+106) = (7+8)+106$ ㉗ $15 \times 3 = 3 \times 15$
- ㉘ $9 = 9 \times 1$ ㉙ $8+3 = 3+8$ ㉚ $(6 \times 3) \times 2 = 6 \times (3 \times 2)$

Divide the numbers below. Write the quotients as mixed numbers.

④ $7 \div 4$

④ $9 \div 2$

④ $8 \div 5$

④ $7 \div 3$

④ $12 \div 5$

④ $78 \div 9$

④ $59 \div 8$

④ $76 \div 9$

④ $52 \div 6$

Convert the rates below to unit rates.

⑤ 51 miles in 9 minutes.

⑤ 26 gallons for 6 yards.

⑤ $144.2 \div 5$

⑤ 84 dollars for every 9 minutes.

⑤ 37 hours for every 8 machines. ⑤ 117 servings for every 7 pounds of dough.

⑤ 830 ounces of water for every 3 grams of sugar.

⑤ 2,133 cups of flour for every 6 teaspoons of vinegar.

⑤ 7 kilograms for every 8 cubic feet ⑤ 4 pounds of meal for 9 dogs.

⑥ 11 miles for every 15 gallons of gas. ⑥ 8 pieces of chocolate for 13 people.